IOCL GRADE-A 2026
CS/IT WORKBOOK

DAY 2 — FUNCTIONAL DEPENDENCIES & NORMALIZATION

20 AUGUST 2026 (Updated Edition)

Topic: Functional Dependencies, Attribute Closure, Candidate Keys, Prime/Non-Prime Attributes, 1NF–3NF, Highest-Normal-Form Determination, Lossless Decomposition & Dependency Preservation (recognition level)

Target session length:	$\approx$ 2 hours 15 minutes
Exam date:	24 September 2026
Study date:	20 August 2026
Follows on from:	Day 1 – DBMS Foundations (19 August 2026)
Leads into:	Day 3 – BCNF, deeper decomposition theory, mixed MCQs

Goal: Comfortable CBT clearance – high-yield Easy/Medium level MCQs, with a smaller number of useful Hard questions. This is *not* GATE-depth mastery, and no unsupported IOCL-specific topic-weightage claims are made anywhere in this workbook.

Session Plan ($\approx$2h15m):

55–60 min	Concepts: FDs, closure, candidate keys, 1NF/2NF/3NF, HNF procedure
25–30 min	Guided practice (closure + key tracing, hints then solutions)
30–35 min	Independent MCQs, normalization problems, Exam Pattern Bank
15–20 min	Timed mini-test (10Q)
10 min	Error log + revision checklist + cheat sheet

This is a study diagnostic tool, not an official IOCL cutoff predictor.
BCNF, 4NF/5NF, Armstrong's axioms, minimal cover, and formal lossless/dependency-preservation proofs are NOT covered today – see Section 3.

Contents

1 Learning Objectives

By the end of this session you should be able to:

- Understand what a functional dependency $X \to Y$ means, and distinguish trivial vs. non-trivial FDs.
- Compute the closure X^+ of a small attribute set using the standard algorithm.
- Use closure to test whether a given attribute set is a super key.
- Identify candidate keys from a small relation and FD set, understanding minimality.
- Distinguish prime vs. non-prime attributes.
- Explain why normalization exists (redundancy, update/insertion/deletion anomalies).
- Distinguish 1NF, 2NF, and 3NF, and identify partial vs. transitive dependency.
- Apply a mechanical step-by-step procedure to determine the *highest normal form* of a relation given its candidate key and FD set.
- Recognize lossless decomposition and dependency preservation at a conceptual/recognition level (no chase-table proofs).
- Confidently attempt PSU/IOCL-pattern FD and normalization questions, and correctly distinguish them from verified GATE PYQs.

Why this matters for IOCL CBT

Normalization theory (FDs, closure, candidate keys, 1NF–3NF) is one of the most frequently and predictably tested DBMS topics in PSU technical sections. Most questions are short closure calculations or definitional traps, not long proofs – ideal for a focused, high-yield session. This updated edition adds a dedicated **Exam Pattern Bank** (Section 7) so you also get practice distinguishing verified GATE-level questions from original PSU-style practice items, and a mechanical **Highest-Normal-Form Decision Procedure** (Section 2.8) so that "which normal form is this relation in?" questions become a fast, repeatable four-step check instead of a judgment call.

2 Concepts and Mental Models

2.1 A. Functional Dependency (FD)

What: A functional dependency $X \to Y$ on relation R means: for any two tuples that agree on all attributes in X, they must also agree on all attributes in Y. X is the **determinant**, Y is the **dependent**.

Why it exists: FDs formally capture real-world business rules (e.g., "RollNo determines Name") so we can reason mathematically about redundancy and design quality.

Trivial vs. non-trivial: $X \to Y$ is **trivial** if $Y \subseteq X$ (always true, e.g. $AB \to A$). It is **non-trivial** if $Y \not\subseteq X$ – these are the FDs that carry real information.

Example FD	Trivial or non-trivial?
RollNo $\to$ Name	Non-trivial (Name $\not\subseteq$ {RollNo})
{RollNo, Name} $\to$ RollNo	Trivial (RHS $\subseteq$ LHS)
$AB \to ABC$	Non-trivial (C is new information)

Common Trap

Repeated or coincidentally matching values in a small sample table do NOT prove an FD holds in general – an FD is a declared/derived business rule, not something inferred purely by eyeballing a few rows.

2.2 B. Attribute Closure (X^+)

What: The closure X^+ of an attribute set X (under a set of FDs F) is the set of all attributes that can be functionally determined starting from X, applying the FDs repeatedly.

Why it matters: If X^+ contains all attributes of R, then X is a **super key** of R. This is the standard mechanical test for super-key-ness.

Mental Model

"Keep adding everything X can reach." Start with X; whenever some FD's LHS is fully covered by what you have, add its RHS; repeat until nothing new is added.

Closure Algorithm

1. Initialize $result = X$.
2. Repeat: for each FD $\alpha \to \beta$ in F, if $\alpha \subseteq result$, add β to $result$.
3. Stop when no new attribute can be added in a full pass.
4. Final $result = X^+$.

Worked closure example

$R(A, B, C, D)$ with FDs: $A \to B$, $B \to C$, $AC \to D$. Compute A^+.
Step 0: $result = \{A\}$. Step 1: $A \to B$ applies $\Rightarrow$ add B. $result = \{A, B\}$. Step 2: $B \to C$ applies $\Rightarrow$ add C. $result = \{A, B, C\}$. Step 3: $AC \to D$ applies ($AC \subseteq result$) $\Rightarrow$ add D. $result = \{A, B, C, D\}$.
No new attributes remain. $A^+ = \{A, B, C, D\}$ = all attributes $\Rightarrow$ A is a super key.

Common Trap

Stopping the closure algorithm too early (after one pass) and missing an FD that only becomes usable once an earlier RHS has been added. Always keep re-scanning all FDs until a full pass adds nothing new.

2.3 C. Candidate Keys

Super key: Any attribute set whose closure equals all attributes of R.

Candidate key: A **minimal** super key – removing any single attribute from it breaks the super-key property (its closure would no longer cover all attributes).

Prime attribute: Any attribute that is part of at least one candidate key. **Non-prime attribute:** An attribute that is part of no candidate key.

Mental Model

"Enough to identify the row, but nothing unnecessary." If removing an attribute still lets you reach everything via closure, that attribute was never needed.

Systematic method for small relations

1. List all attributes that never appear on the RHS of any FD – these must belong to *every* candidate key (call this the "core").
2. Compute the closure of the core. If it covers all attributes, the core itself is a candidate key.
3. If not, try adding one attribute at a time (smallest sets first) and recompute closure until you find minimal sets that reach all attributes.
4. Check minimality: for each candidate found, confirm no single attribute can be removed without losing full closure.

Worked example 1 – single-attribute candidate key

$R(A, B, C, D)$, FDs: $A \to B$, $B \to C$, $AC \to D$. A never appears on the RHS of any FD $\Rightarrow$ A is in the core. $A^+ = \{A, B, C, D\}$ = all attributes $\Rightarrow$ $\{A\}$ is a candidate key (single attribute $\Rightarrow$ automatically minimal). Prime attribute: A. Non-prime: B, C, D.

Worked example 2 – composite candidate key, multiple keys

$R(P, Q, R)$, FDs: $PQ \to R$, $R \to Q$. Neither P nor Q alone reaches everything ($P^+ = \{P\}$; $Q^+ = \{Q\}$). Try $\{P, R\}$: $R \to Q$ gives $\{P, R, Q\}$ = all $\Rightarrow$ super key; minimal ($\{P\}^+ = \{P\}$ fails, $\{R\}^+ = \{R, Q\}$ fails, no P) $\Rightarrow$ candidate key. Also $\{P, Q\}$: $PQ \to R$ gives all $\Rightarrow$ super key; minimal too $\Rightarrow$ candidate key. **This relation has two candidate keys: $\{P, Q\}$ and $\{P, R\}$.** Prime attributes: P, Q, R (all prime – each appears in some candidate key).

Worked example 3 – a super key that is NOT minimal

Same $R(A, B, C, D)$ as example 1. Consider $\{A, B\}$: closure is $\{A, B, C, D\}$ = all attributes, so $\{A, B\}$ IS a super key. But is it minimal? Remove B: $\{A\}^+ = \{A, B, C, D\}$ – still all attributes! B was redundant $\Rightarrow$ $\{A, B\}$ is a super key but NOT a candidate key.

Common Trap

A relation can have multiple candidate keys (Worked Example 2). Also, just because a set's closure covers all attributes does not automatically make it a candidate key – minimality must also be verified (Worked Example 3).

2.4 D. Why Normalization Exists

The three anomalies. Unnormalized/poorly designed tables that mix unrelated facts in one relation cause:

- **Redundancy:** the same fact stored in multiple rows.
- **Update anomaly:** changing one instance of a repeated fact requires updating many rows; missing even one causes inconsistency.
- **Insertion anomaly:** you cannot insert some information without also having unrelated information available.
- **Deletion anomaly:** deleting one fact unintentionally deletes other, unrelated information.

RollNo	Name	CourseID	CourseName
101	Asha	C1	DBMS
101	Asha	C2	OS
102	Bala	C1	DBMS

Redundancy: "Asha" repeated. *Update anomaly:* renaming C1 requires updating 2 rows. *Insertion anomaly:* cannot add course C3 unless some student enrolls. *Deletion anomaly:* if Bala is the only student in C1 and we delete his row, we lose all record of C1 existing.

2.5 E. First Normal Form (1NF)

Definition: A relation is in 1NF if every attribute holds only atomic (indivisible) values – no repeating groups, no multi-valued attributes, no nested tables within a single cell.

Counterexample (not in 1NF):

RollNo	Name	Phones
101	Asha	98765xxxxx, 91234xxxxx

The Phones column holds multiple values in one cell – violates 1NF. Fix: split into separate rows $(RollNo, Phone)$ or a separate related table.

Common Trap

1NF guarantees atomicity ONLY – it does not remove redundancy, partial dependency, or transitive dependency. A table can be in 1NF and still suffer from all the anomalies described above.

2.6 F. Second Normal Form (2NF)

Definition: A relation is in 2NF if it is already in 1NF, AND no non-prime attribute is **partially dependent** on any candidate key – i.e., no non-prime attribute depends on only a proper subset of a composite candidate key.

Mental Model

"No dependency on only part of a composite key." 2NF issues can ONLY occur when at least one candidate key is composite (2+ attributes). If every candidate key is a single attribute, 2NF is automatically satisfied once 1NF holds.

Worked example – 2NF violation

$R(StudentID, CourseID, StudentName, Marks)$, candidate key $= \{StudentID, CourseID\}$. FD: $StudentID \to StudentName$ (depends on only part of the key). This is a **partial dependency** $\Rightarrow$ violates 2NF. Fix: decompose into $R_1(StudentID, StudentName)$ and $R_2(StudentID, CourseID, Marks)$.

Common Trap

2NF is about *any* candidate key being composite, and specifically about *non-prime* attributes – a prime attribute depending on part of a composite key does NOT violate 2NF. Also: 2NF is a concern for every composite candidate key the relation has, not just the one column set chosen as the "primary" key.

2.7 G. Third Normal Form (3NF)

Practical definition: A relation is in 3NF if it is in 2NF, AND no non-prime attribute is **transitively dependent** on a candidate key through another non-prime attribute.

Formal condition (recognition level): For every non-trivial FD $X \to A$, either X is a super key, OR A is a prime attribute.

Mental Model

"No non-key attribute depending on another non-key attribute through the key." The chain $Key \to NonKey_1 \to NonKey_2$ is the red flag.

Worked example – 2NF satisfied but 3NF violated

$R(EmpID, DeptID, DeptName)$, candidate key $= \{EmpID\}$ (single attribute, so 2NF holds automatically). FDs: $EmpID \to DeptID$, $DeptID \to DeptName$. $EmpID \to DeptName$ holds transitively via the non-prime attribute $DeptID \Rightarrow$ violates 3NF. Fix: decompose into $R_1(EmpID, DeptID)$ and $R_2(DeptID, DeptName)$.

Common Trap

Trap 1: Confusing partial dependency (2NF issue – depends on part of a composite key) with transitive dependency (3NF issue – depends on a non-key attribute). **Trap 2:** "3NF just means no redundancy" is an oversimplification – 3NF specifically targets transitive dependencies through non-prime attributes; some redundancy can still exist even in 3NF (fully removed only at BCNF, not covered today – see Section 3).

2.8 H. Highest-Normal-Form Decision Procedure

Many CBT questions give you a relation, its candidate key, and a small FD set, then ask "what is the highest normal form this relation satisfies?" The following four-step procedure answers this mechanically, without needing to reason from scratch each time.

Mechanical Exam Procedure

Step 1 – Check atomicity → 1NF. If any cell can hold more than one value, the relation fails 1NF and no higher normal form applies. Otherwise 1NF holds; continue.
Step 2 – Identify candidate key(s); mark prime/non-prime attributes. Use the closure method from Section 2.3.
Step 3 – Check partial dependency → 2NF. For every non-prime attribute, does it depend on a *proper subset* of some composite candidate key? If yes, 2NF fails (relation is in 1NF only). If every candidate key is single-attribute, or no such dependency exists, 2NF holds; continue.
Step 4 – Check transitive dependency → 3NF. For every non-trivial FD $X \to A$ where A is non-prime, is X a super key? If some non-prime A depends on a non-superkey X (equivalently: depends via another non-prime attribute), 3NF fails (relation is in 2NF only). Otherwise 3NF holds.

Worked example

$R(A, B, C, D)$, candidate key $= AB$, FDs $= \{AB \to C,\ A \to D\}$.
Step 1: Values assumed atomic ⇒ 1NF holds.
Step 2: Candidate key $= AB$ (given). Prime: A, B. Non-prime: C, D.
Step 3: Check $A \to D$: A is a proper subset of the composite key AB, and D is non-prime ⇒ **partial dependency ⇒ 2NF is violated. The relation is in 1NF only.**
Violating FD: $A \to D$.
Fix: Decompose into $R_1(A, D)$ (captures $A \to D$) and $R_2(A, B, C)$ (captures $AB \to C$, with A, B still the key of R_2). Check: in R_2, key $= AB$ and C depends on the full key – no partial dependency. In R_1, the key is the single attribute A – no composite key, so no partial dependency is possible. The violation is removed.

Common Trap

Trap 1: 2NF is about *every* candidate key, not merely the one column set chosen as the primary key. **Trap 2:** 2NF is only ever a concern when a composite candidate key exists – with an all-single-attribute-key relation, jump straight from 1NF to checking 3NF. **Trap 3:** A *prime* attribute depending on part of a composite key does NOT violate 2NF – only non-prime attributes count. **Trap 4:** "3NF" is not simply "no redundancy" – use the formal condition: for every non-trivial FD $X \to A$, either X is a super key OR A is prime.

2.9 I. Lossless Decomposition and Dependency Preservation

Definitions – recognition level only.

Lossless decomposition: Splitting R into R_1, R_2 is lossless if joining R_1 and R_2 back together (natural join) reproduces exactly the original relation R – no spurious extra rows and no lost rows. This requires the common attribute(s) between R_1 and R_2 to form a key in at least one of them.

Dependency preservation: A decomposition preserves dependencies if all the original FDs can still be logically enforced using only the FDs that hold within the individual decomposed tables, without needing to rejoin them.

Mental Model

Losslessness asks: "If I rejoin the pieces, do I get back exactly what I started with?"
Dependency preservation asks: "Can I still enforce all the original business rules using just

the separate pieces, without having to join them first?" **These are two independent properties** – a decomposition can be lossless but not dependency-preserving, or vice versa.

Common Trap

Assuming every decomposition performed for normalization purposes is automatically lossless – it is NOT automatic; it must be verified (the common attribute must be a key in one of the resulting tables). Do not confuse "lossless" with "dependency-preserving" – they are separate, independently-checked properties.

3 What Comes Later — Do Not Study Now

These are valid and important DBMS topics, but they are **intentionally postponed** to later study days. Studying them now will not help your Day 2 CBT score and will eat into time better spent mastering closure, candidate keys, and 1NF–3NF cold.

- Boyce-Codd Normal Form (BCNF)
- Armstrong's axioms and derived inference rules (union, decomposition, pseudo-transitivity)
- Minimal / canonical cover
- Equivalence of FD sets
- Full lossless-join proofs / the chase algorithm
- Formal dependency-preservation algorithms and proofs
- Fourth Normal Form (4NF) and multivalued dependencies
- Fifth Normal Form (5NF) and join dependencies
- Advanced decomposition theory beyond the recognition-level checks in Section 2.9

Common Trap

If a question in a mock test asks about BCNF, minimal cover, or a chase-table proof before you reach Day 3, do not panic and do not stop to learn it mid-session – flag it, guess your best answer using what you know from 3NF, and move on. These topics get their own dedicated session on Day 3.

4 Guided Practice (10 Questions – hints, then full solutions)

Attempt each question, read the hint if stuck, then check the full worked solution in Section 5 (Guided Practice – Full Worked Solutions) before moving to Independent Practice.

4.1 FD / Closure (3 questions)

G1 [E] [FD][Trivial/non-trivial] $R(A, B, C, D)$ with FD $ABC \to AB$. Is this trivial or non-trivial?

Hint

Check whether the RHS is a subset of the LHS.

G2 [M] [Closure] $R(A, B, C, D, E)$ with FDs: $A \to B$, $AB \to C$, $D \to E$. Compute A^+. Is $\{A\}$ a super key of R? Which attribute can never be reached from A alone?

Hint

Add B first, then check whether $AB \to C$ now applies. Does anything ever put D or E within reach of A alone?

G3 [M] [Closure] $R(P, Q, R, S, T)$ with FDs: $P \to Q$, $Q \to R$, $R \to S$. Compute P^+. Is $\{P\}$ a super key?

Hint

This is a simple chain – follow $P \to Q \to R \to S$ one step at a time. Does any FD ever produce T?

4.2 Candidate Keys (3 questions)

G4 [M] [Candidate Key] $R(A, B, C, D)$ with FDs: $A \to B$, $B \to C$, $AC \to D$. Confirm $\{A\}$ is a candidate key, and state why $\{A, B\}$ is NOT one.

Hint

This matches Worked Example 1 / Worked Example 3 in Section 2.3 – re-derive minimality for $\{A, B\}$.

G5 [M] [Candidate Key] $R(P, Q, R)$ with FDs: $PQ \to R$, $R \to Q$. List all candidate keys.

Hint

This matches Worked Example 2 – try $\{P, Q\}$ and $\{P, R\}$. Is $\{P\}$ alone ever enough?

G6 [H] [Candidate Key + Prime Attributes] $R(A, B, C, D, E)$ with FDs: $AB \to C$, $C \to D$, $D \to E$. Is $\{A, B\}$ a candidate key? Which attributes are prime?

Hint

Compute $\{A, B\}^+$ fully first, then check whether removing A or removing B individually still reaches all attributes.

4.3 Normalization (4 questions)

G7 [E] [1NF] A table has a column storing comma-separated multiple email addresses in a single cell. Which normal form does this violate?

Hint

Think about atomicity of attribute values.

G8 [M] [2NF] $R(OrderID, ProductID, ProductName, Qty)$, candidate key = $\{OrderID, ProductID\}$. FD: $ProductID \to ProductName$. Which normal form is violated, and why?

Hint

Does $ProductName$ depend on the whole composite key, or just part of it?

G9 [M] [3NF] $R(EmpID, City, State)$, candidate key = $\{EmpID\}$. FDs: $EmpID \to City$, $City \to State$. Which normal form is violated, and why?

Hint

Is $State$ depending on $EmpID$ directly, or via another non-prime attribute?

G10 [M] [Highest Normal Form] $R(A, B, C, D)$, candidate key = AB, FDs = $\{AB \to C,\ A \to D\}$. Is R in 1NF? In 2NF? In 3NF? Which FD causes the violation, and what decomposition removes it?

Hint

Apply the four-step procedure from Section 2.8 directly.

5 Guided Practice – Full Worked Solutions

G1 Solution

$ABC \to AB$: RHS $= \{A, B\}$, LHS $= \{A, B, C\}$. RHS $\subseteq$ LHS $\Rightarrow$ **trivial**.

G2 Solution

A^+: start $\{A\}$. $A \to B$: add $B \Rightarrow \{A, B\}$. Now $AB \to C$ applies ($AB \subseteq$ result): add $C \Rightarrow \{A, B, C\}$. $D \to E$ never fires (D is not yet in the result, and nothing ever puts it there). $A^+ = \{A, B, C\}$, NOT all attributes (missing D, E) $\Rightarrow \{A\}$ **is NOT a super key.** D (and hence E) can never be reached from A alone, since nothing determines D.

G3 Solution

P^+: $\{P\} \to$ add Q (from $P \to Q$) $\to \{P, Q\} \to$ add R (from $Q \to R$) $\to \{P, Q, R\} \to$ add S (from $R \to S$) $\to \{P, Q, R, S\}$. No FD ever produces T. $P^+ = \{P, Q, R, S\} \neq$ all attributes $\Rightarrow \{P\}$ **is NOT a super key.**

G4 Solution

$\{A\}$: $A^+ = \{A, B, C, D\}$ (via $A \to B$, $B \to C$, $AC \to D$) = all attributes $\Rightarrow$ super key; single attribute so automatically minimal $\Rightarrow$ **candidate key**. For $\{A, B\}$: closure is also $\{A, B, C, D\}$ (super key), but removing B gives $\{A\}^+ = \{A, B, C, D\}$ – still all attributes, so B was redundant. $\{A, B\}$ **fails minimality** $\Rightarrow$ not a candidate key (it is a non-minimal super key).

G5 Solution

$\{P\}$ alone: $P^+ = \{P\}$ – not a super key. $\{P, Q\}$: $PQ \to R$ gives $\{P, Q, R\}$ = all $\Rightarrow$ super key; minimal since removing either P or Q loses closure. $\{P, R\}$: $R \to Q$ gives $\{P, R, Q\}$ = all $\Rightarrow$ super key; minimal similarly. **Candidate keys:** $\{P, Q\}$ **and** $\{P, R\}$ (this relation has two candidate keys).

G6 Solution

$\{A, B\}^+$: start $\{A, B\} \to AB \to C$ adds $C \to \{A, B, C\} \to C \to D$ adds $D \to \{A, B, C, D\} \to D \to E$ adds $E \to \{A, B, C, D, E\}$ = all attributes $\Rightarrow$ super key. Minimality: $\{A\}^+ = \{A\}$ (fails), $\{B\}^+ = \{B\}$ (fails) $\Rightarrow \{A, B\}$ **is a candidate key.** (Since A, B never appear on any RHS, this is also the only candidate key.) **Prime attributes:** A, B**. Non-prime:** C, D, E**.**

G7 Solution

A single cell holding multiple comma-separated values is a multi-valued, non-atomic attribute $\Rightarrow$ **violates 1NF.**

G8 Solution

$ProductName$ depends on $ProductID$ alone, which is only part of the composite candidate key $\{OrderID, ProductID\} \Rightarrow$ this is a **partial dependency** $\Rightarrow$ **violates 2NF** (assuming 1NF already holds).

G9 Solution

$EmpID \to City \to State$: $State$ depends on $EmpID$ only transitively, through the non-prime attribute $City \Rightarrow$ **violates 3NF** (2NF holds trivially since the candidate key $\{EmpID\}$ is a single attribute, so no partial dependency is possible).

G10 Solution

Step 1: Atomic values assumed $\Rightarrow$ 1NF holds. *Step 2:* Candidate key $= AB$ (given); prime $= A, B$; non-prime $= C, D$. *Step 3:* Check $A \to D$: A is a proper subset of the composite key AB, and D is non-prime $\Rightarrow$ **partial dependency** $\Rightarrow$ **2NF is violated.** R **is in 1NF only. Violating FD:** $A \to D$. **Fix:** decompose into $R_1(A, D)$ and $R_2(A, B, C)$ – both pieces are now free of partial dependency (verify: R_2's only non-trivial FD $AB \to C$ depends on the full key; R_1 has a single-attribute key, so no composite-key issue is possible).

6 Independent Practice (16 Questions)

Attempt without referring back. Distribution: 5 FD/closure, 4 candidate-key, 5 normalization (including one highest-normal-form question), 2 lossless/dependency-preservation recognition. Answers appear only in the Answer Key (Section 10).

6.1 Functional Dependency / Closure (5 questions)

I1 [E] [FD][Concept] $X \to Y$ is called trivial when:
(A) $X \subseteq Y$ (B) $Y \subseteq X$ (C) $X \cap Y = \emptyset$ (D) $X = \emptyset$

I2 [E] [FD][Concept] In the FD $CustomerID \to CustomerName$, the determinant is:
(A) $CustomerName$ (B) $CustomerID$ (C) Both (D) Neither

I3 [M] [Closure][Problem] $R(A, B, C, D)$, FD: $A \to BC$. What is A^+?
(A) $\{A\}$ (B) $\{A, B\}$ (C) $\{A, B, C\}$ (D) $\{A, B, C, D\}$

I4 [M] [Closure][Problem] $R(A, B, C, D)$, FDs: $AB \to C$, $C \to D$. What is $\{A, B\}^+$?
(A) $\{A, B\}$ (B) $\{A, B, C\}$ (C) $\{A, B, C, D\}$ (D) $\{C, D\}$

I5 [H] [Closure][Trap] $R(A, B, C, D, E)$, FDs: $A \to B$, $BC \to D$, $D \to E$. Compute $(AC)^+$. Is $\{A, C\}$ a candidate key?
(A) $(AC)^+ = \{A, B, C\}$; not a candidate key (B) $(AC)^+ = \{A, B, C, D, E\}$; it is a candidate key (C) $(AC)^+ = \{A, B, C, D, E\}$; it is a super key but not minimal (D) $(AC)^+ = \{A, C\}$; not a super key

6.2 Candidate Keys (4 questions)

I6 [E] [Candidate Key][Definition] A super key that cannot lose any attribute without losing closure over all attributes is called a:
(A) Foreign key (B) Candidate key (C) Alternate key (D) Composite key

I7 [E] [Candidate Key][Definition] An attribute that belongs to no candidate key is called:
(A) Prime (B) Non-prime (C) Composite (D) Trivial

I8 [M] [Candidate Key][Problem] $R(A, B, C)$, FDs: $A \to B$, $B \to A$, $AB \to C$. Which of these is a candidate key?
(A) $\{C\}$ (B) $\{A\}$ (C) $\{B, C\}$ (D) Both $\{A\}$ and $\{B\}$

I9 [H] [Candidate Key][Trap] $R(A, B, C, D)$, FDs: $A \to BC$, $C \to D$. A student claims $\{A, C\}$ is the only candidate key. What's wrong with this claim?
(A) Nothing, it's correct (B) $\{A\}$ alone is already a candidate key, making $\{A, C\}$ non-minimal
(C) $\{A, C\}$ is not a super key (D) D can never be determined

6.3 Normalization (5 questions)

I10 [E] [Normalization][Definition] A relation is in 1NF if:
(A) It has no candidate keys (B) All attribute values are atomic (C) It has no foreign keys
(D) It has only one candidate key

I11 [E] [Normalization][Definition] A partial dependency can only occur when:
(A) The candidate key is a single attribute (B) The candidate key is composite (C) There are no foreign keys (D) The relation is in BCNF

I12 [M] [Normalization][Scenario] $R(StudentID, CourseID, Grade)$ with candidate key $\{StudentID, CourseID\}$ and no other FDs among non-key attributes. This relation is in:
(A) 1NF only (B) 2NF but not 3NF (C) 3NF (D) None of the normal forms

I13 [M] [Normalization][Trap] Which statement about 3NF is TRUE?

(A) 3NF eliminates all possible redundancy (B) 3NF requires the relation to already be in 2NF (C) 3NF is only relevant for composite keys (D) 3NF and 1NF are unrelated

I14 [M] [Highest Normal Form][Problem] $R(A, B, C, D)$, candidate key = $\{A\}$ (single attribute), FDs: $A \to B$, $B \to C$, $A \to D$. What is the highest normal form R satisfies?
(A) 1NF only (B) 2NF but not 3NF (C) 3NF (D) Cannot be determined

6.4 Lossless / Dependency-Preservation Recognition (2 questions)

I15 [M] [Decomposition][Concept] A decomposition of R into R_1 and R_2 is lossless when:
(A) R_1 and R_2 have the same number of attributes (B) Natural-joining R_1 and R_2 reproduces exactly R (C) R_1 and R_2 share no attributes (D) Every FD of R appears in R_1

I16 [E] [Decomposition][Concept] Dependency preservation and losslessness are:
(A) The same property (B) Independent properties – one can hold without the other (C) Mutually exclusive (never both true) (D) Only relevant in BCNF

7 Exam Pattern Bank

Common Trap

Source labeling in this bank: Questions marked **[Verified GATE PYQ – ...]** are real, previously administered GATE questions, checked against publicly documented GATE archives and solved independently in this workbook to confirm the answer. Questions marked **[PSU/IOCL-pattern practice]** are **original** practice questions written in the style commonly seen in PSU/IOCL-type papers – they are **not** claimed to be actual IOCL previous-year questions, since no IOCL-specific FD/normalization PYQ could be independently verified from a public source at the time of writing. No claim is made about how often any of these patterns appear specifically in IOCL's paper.

V1 [Verified GATE PYQ – GATE (IT) 2005, Q5] [Closure, Highest Normal Form] [M]

A table has fields F_1, F_2, F_3, F_4, F_5 with functional dependencies: $F_1 \to F_3$, $F_2 \to F_4$, $(F_1, F_2) \to F_5$. In terms of normalization, this table is in:
(A) 1NF (B) 2NF (C) 3NF (D) None of these

V2 [Verified GATE PYQ – GATE CS 2013] [Candidate-Key Finding, Highest Normal Form] [H]

$R(A, B, C, D, E, F, G, H)$, each field holds only atomic values, and $F = \{CH \to G,\ A \to BC,\ B \to CFH,\ E \to A,\ F \to EG\}$ is exactly the set of FDs that holds on R. Find all candidate keys of R, and state whether R is in 2NF.

V3 [Verified GATE PYQ – GATE CSE 2022, NAT] [Closure, Candidate Key] [M]

$R(A, B, C, D, E)$ with FDs: $AB \to C$, $BC \to D$, $C \to E$. Find the candidate key(s) of R, and state how many super keys R has.

P1 [PSU/IOCL-pattern practice] [FD][Trivial/Non-trivial] [E]

Is $\{P, Q, R\} \to Q$ trivial or non-trivial?

P2 [PSU/IOCL-pattern practice] [Closure] [E]

$R(A, B, C, D)$, FDs: $A \to B$, $A \to C$. Find A^+. Is $\{A\}$ a super key?

P3 [PSU/IOCL-pattern practice] [Candidate Key] [M]

$R(A, B, C, D)$, FDs: $AB \to CD$, $CD \to AB$. Find all candidate keys, and list the prime attributes.

P4 [PSU/IOCL-pattern practice] [Prime/Non-prime] [E]

$R(A, B, C, D, E)$, candidate keys $\{A, B\}$ and $\{C\}$. List the prime and non-prime attributes.

P5 [PSU/IOCL-pattern practice] [Highest Normal Form][Trap] [H]

$R(RollNo, Subject, Marks, Grade)$, candidate key $= \{RollNo, Subject\}$. FD: $Marks \to Grade$. Determine the highest normal form of R, and be precise about whether this is a partial dependency or a transitive dependency.

P6 [PSU/IOCL-pattern practice] [Lossless Decomposition Recognition] [M]

$R(A, B, C)$ is decomposed into $R_1(A, B)$ and $R_2(B, C)$. Given that FD $B \to A$ holds on R, is this decomposition lossless?

P7 [PSU/IOCL-pattern practice] [Dependency-Preservation Recognition] [M]

$R(EmpID, DeptID, DeptName)$, FDs: $EmpID \to DeptID$, $DeptID \to DeptName$, is decomposed into $R_1(EmpID, DeptID)$ and $R_2(DeptID, DeptName)$. Is this decomposition lossless? Is it dependency-preserving?

Exam Pattern Bank – Complete Solutions

V1 Solution

Since the primary key is not stated, derive it: F_1 alone reaches only F_3; F_2 alone reaches only F_4; neither alone reaches everything. $(F_1, F_2)^+$: add F_3 (via $F_1 \to F_3$), add F_4 (via $F_2 \to F_4$), add F_5 (via $(F_1, F_2) \to F_5$) $= \{F_1, \ldots, F_5\} = R$. So $\{F_1, F_2\}$ is the (minimal) candidate key. Non-prime: F_3, F_4, F_5. Now check 2NF: $F_1 \to F_3$ – F_1 is a proper subset of the composite key $\{F_1, F_2\}$, and F_3 is non-prime ⇒ **partial dependency**. Likewise $F_2 \to F_4$ is also partial. **Answer: (A) 1NF** (2NF is violated).

V2 Solution

RHS attributes across F: G, B, C, F, H, A, E – every attribute except D. So D must be in every candidate key. Testing D combined with each other attribute: $(AD)^+$: $A \to BC$ adds $B, C \to \{A, B, C, D\}$; $B \to CFH$ adds $F, H \to \{A, B, C, D, F, H\}$; $F \to EG$ adds $E, G \to$ all 8 attributes. So AD is a candidate key. By the same style of expansion, BD, ED, and FD each also reach all 8 attributes and are minimal (verify: none of A, B, E, F alone reaches D, since nothing determines D). CD does *not* work: C alone contributes nothing without H (the FD $CH \to G$ needs both), so $\{C, D\}^+ = \{C, D\}$ only – not a super key. **Candidate keys:** AD, BD, ED, FD**.** Prime attributes: A, B, D, E, F. Non-prime: C, G, H. Checking 2NF: $A \to BC$ – A is a proper subset of the composite key AD, and C is non-prime ⇒ **partial dependency** ⇒ R **is in 1NF only, not 2NF.**

V3 Solution

RHS attributes across F: C, D, E. So A, B never appear on any RHS and must be in every candidate key. $(AB)^+$: $AB \to C$ adds $C \to \{A, B, C\}$; $BC \to D$ adds $D \to \{A, B, C, D\}$; $C \to E$ adds $E \to \{A, B, C, D, E\} = R$. So AB is a super key; minimal since $A^+ = \{A\} \neq R$ and $B^+ = \{B\} \neq R$. AB **is the only candidate key** ($n = 5$, $k = 2$). **Number of super keys** $= 2^{n-k} = 2^{5-2} = \mathbf{8}$.

P1 Solution

$Q \subseteq \{P, Q, R\}$, so RHS $\subseteq$ LHS ⇒ **trivial**.

P2 Solution

A^+: $A \to B$ adds B; $A \to C$ adds C. $A^+ = \{A, B, C\}$. Not all attributes (D missing) ⇒ $\{A\}$ **is NOT a super key.**

P3 Solution

$(AB)^+$: $AB \to CD$ gives $\{A, B, C, D\} = R$ ⇒ super key; minimal ($A^+ = \{A\} \neq R$, $B^+ = \{B\} \neq R$) ⇒ **candidate key**. $(CD)^+$: $CD \to AB$ gives $\{A, B, C, D\} = R$ ⇒ super key; minimal similarly ⇒ **candidate key**. **Candidate keys:** AB **and** CD**.** Since every attribute belongs to one of these two keys, **all four attributes (**A, B, C, D**) are prime** – there are no non-prime attributes here.

P4 Solution

Prime: A, B, C (each is part of a candidate key). Non-prime: D, E.

P5 Solution

Non-prime attributes: $Marks, Grade$ (neither is part of the candidate key $\{RollNo, Subject\}$). Check the FD $Marks \to Grade$: is $Marks$ a *proper subset* of the candidate key $\{RollNo, Subject\}$? **No** – $Marks$ is not part of the key at all, so this is **not** a partial dependency (2NF is specifically about a non-prime attribute depending on part of a composite key; here the determinant isn't part of the key). Instead, $Marks$ is a non-prime attribute that is itself not a super key ($Marks^+ = \{Marks, Grade\} \neq R$), and it determines another non-prime attribute $Grade$ – this is a **transitive dependency** $\Rightarrow$ **violates 3NF**. R **is in 2NF but not 3NF.** Fix: decompose into $R_1(RollNo, Subject, Marks)$ and $R_2(Marks, Grade)$.

P6 Solution

Common attribute between $R_1(A, B)$ and $R_2(B, C)$ is B. Losslessness requires B to be a key of R_1 or R_2. Since $B \to A$ holds and B is a single attribute, B is a candidate key of $R_1(A, B) \Rightarrow$ **the decomposition is lossless.**

P7 Solution

Common attribute is $DeptID$. $DeptID \to DeptName$ holds and $DeptID$ is a single attribute, so $DeptID$ is a candidate key of $R_2 \Rightarrow$ **lossless.** Dependency preservation: $EmpID \to DeptID$ holds fully within R_1, and $DeptID \to DeptName$ holds fully within R_2 – both original FDs are enforceable without rejoining $\Rightarrow$ **dependency-preserving too.**

8 Timed Mini-Test (10 Questions – 15 minutes)

Instructions: Set a timer for 15 minutes. Attempt all 10 without pausing. Answers appear only in the Answer Key (Section 10) – do not look ahead.

T1 [E] [FD][Definition] $A \to A$ is:
(A) Trivial (B) Non-trivial (C) Undefined (D) A candidate key

T2 [M] [Closure][Problem] $R(A, B, C, D)$, FD: $AB \to CD$. What is $(AB)^+$?
(A) $\{A, B\}$ (B) $\{A, B, C\}$ (C) $\{A, B, C, D\}$ (D) $\{C, D\}$

T3 [M] [Closure][Application] If X^+ equals all attributes of R, then X is guaranteed to be a:
(A) Candidate key (B) Super key (C) Foreign key (D) Prime attribute

T4 [E] [Candidate Key][Concept] A relation can have:
(A) Only one candidate key, always (B) Multiple candidate keys (C) No candidate keys, ever (D) A candidate key only if BCNF holds

T5 [M] [Candidate Key][Definition] A prime attribute is one that:
(A) Appears in every relation (B) Belongs to at least one candidate key (C) Is always the primary key (D) Cannot be part of a composite key

T6 [E] [Normalization][Definition] 1NF specifically enforces:
(A) No transitive dependency (B) No partial dependency (C) Atomic attribute values (D) No redundancy at all

T7 [M] [Normalization][Trap] True or False: "2NF violations can occur even when the candidate key is a single attribute."
(A) True (B) False

T8 [M] [Normalization][Formal Condition] The formal 3NF condition states that for every non-trivial FD $X \to A$:
(A) X must be prime (B) X is a super key OR A is prime (C) A must be a super key (D) X and A must both be prime

T9 [M] [Highest Normal Form][Problem] $R(A, B, C)$, candidate key $= \{A\}$, FD: $B \to C$ (both B, C non-prime). What is the highest normal form R satisfies?
(A) 1NF only (B) 2NF but not 3NF (C) 3NF (D) Cannot be determined

T10 [M] [Decomposition][Trap] True or False: "A decomposition that is lossless is automatically dependency-preserving."
(A) True (B) False

9 Daily Quiz (5 Questions)

Final quick check – no solutions immediately after; see Answer Key (Section 10).

Q1 [E] [FD][Concept] $AB \to B$ is:
(A) Trivial (B) Non-trivial (C) Undefined (D) A candidate key

Q2 [M] [Closure][Problem] $R(A, B, C)$, FD: $A \to BC$ only. What is A^+?
(A) $\{A\}$ (B) $\{A, B\}$ (C) $\{A, B, C\}$ (D) $\{B, C\}$

Q3 [E] [Candidate Key][Definition] A prime attribute is one that:
(A) Appears in every relation (B) Belongs to at least one candidate key (C) Is always the primary key (D) Cannot be part of a composite key

Q4 [M] [Normalization][Scenario] $R(EmpID, Skill)$ where an employee's multiple skills are stored as one comma-separated string per employee. This violates:
(A) 2NF (B) 3NF (C) 1NF (D) No normal form is violated

Q5 [M] [Decomposition][Concept] Which of these is true about dependency preservation?
(A) It guarantees losslessness too (B) It means original FDs can be enforced from the decomposed tables without rejoining (C) It only applies to BCNF (D) It is the same as atomicity

10 Complete Answer Key

10.1 Independent Practice – Answers

I1. (B) $Y \subseteq X$**.** Triviality means the RHS is already contained in the LHS.

I2. (B) CustomerID. The determinant is always the LHS of the arrow.

I3. (C) $\{A, B, C\}$**.** $A \to BC$ adds both B and C directly; nothing reaches D.

I4. (C) $\{A, B, C, D\}$**.** $AB \to C$ adds C, then $C \to D$ adds D.

I5. (B) $(AC)^+ = \{A, B, C, D, E\}$**; it is a candidate key.** $A \to B$ adds B; now $BC \to D$ fires (both B, C present) adding D; then $D \to E$ adds E. Full closure reached; $A^+ = \{A, B\} \neq R$ and $C^+ = \{C\} \neq R$ individually, so AC is minimal.

I6. (B) Candidate key. Textbook definition – minimal super key.

I7. (B) Non-prime. Prime = in at least one candidate key; non-prime = in none.

I8. (D) Both $\{A\}$ **and** $\{B\}$**.** $A \to B$ and $B \to A$ mean A, B determine each other; combined with $AB \to C$, either alone reaches C. Both are minimal single-attribute candidate keys.

I9. (B). $A^+ = \{A, B, C, D\}$ already ($A \to BC$ gives B, C; then $C \to D$ gives D), so $\{A\}$ alone is already a candidate key; adding C makes $\{A, C\}$ a non-minimal super key.

I10. (B) All attribute values are atomic. The sole requirement of 1NF.

I11. (B) The candidate key is composite. Partial dependency requires a proper subset of a composite key – impossible with a single-attribute key.

I12. (C) 3NF. With no FDs among the non-key attributes, there is no partial or transitive dependency to violate – the relation satisfies 3NF vacuously.

I13. (B) 3NF requires the relation to already be in 2NF. The standard layered definition. (A) is false – 3NF does not eliminate all redundancy (BCNF goes further); (C) is false – the transitive-dependency rule applies regardless of key shape; (D) is false – normal forms are cumulative.

I14. (B) 2NF but not 3NF. Single-attribute key $\Rightarrow$ 2NF holds automatically. Check 3NF: $B \to C$, where B is non-prime and not a super key ($B^+ = \{B, C\} \neq R$) $\Rightarrow$ transitive dependency $\Rightarrow$ 3NF violated.

I15. (B). Natural-joining the decomposed parts must reproduce the original relation exactly – the definition of losslessness.

I16. (B) Independent properties. A decomposition can be lossless without preserving dependencies, and vice versa.

10.2 Timed Mini-Test – Answers

T1. (A) Trivial. $A \subseteq \{A\}$, so the RHS is a subset of the LHS.

T2. (C) $\{A, B, C, D\}$**.** $AB \to CD$ adds both C and D in one step.

T3. (B) Super key. Full closure guarantees only super-key status; minimality (candidate key) must be checked separately.

T4. (B) Multiple candidate keys. As shown in Worked Example 2 / Guided G5, this is common.

T5. (B). Direct definitional recall: prime = belongs to at least one candidate key.

T6. (C) Atomic attribute values. 1NF's sole job is enforcing atomicity.

T7. (B) False. 2NF violations require a *composite* candidate key – with a single-attribute key, 2NF holds automatically once 1NF holds.

T8. (B). For every non-trivial FD $X \to A$: X is a super key OR A is prime.

T9. (B) 2NF but not 3NF. Single-attribute key $\Rightarrow$ 2NF automatic. $B \to C$: B is non-prime, not a super key ($B^+ = \{B, C\} \neq R$) $\Rightarrow$ transitive dependency $\Rightarrow$ 3NF violated.

T10. (B) False. Losslessness and dependency preservation are independent properties; one does not guarantee the other.

10.3 Daily Quiz – Answers

Q1. (A) Trivial. $B \subseteq \{A, B\}$, so the RHS is a subset of the LHS.

Q2. (C) $\{A, B, C\}$. $A \to BC$ directly adds both B and C in one step.

Q3. (B). Direct definitional recall: prime = belongs to at least one candidate key.

Q4. (C) 1NF. A comma-separated multi-valued cell is a non-atomic value – this is specifically a 1NF violation, occurring before 2NF/3NF even become relevant.

Q5. (B). Dependency preservation means the original FDs can be checked/enforced using only the decomposed tables' own FDs, without needing to rejoin them. It does not guarantee losslessness (a separate property).

11 Scoring and Diagnostic Interpretation

Score band	Interpretation and next action
<60%	Revisit closure and candidate-key worked examples tonight; redo Guided Practice G2–G6 and the HNF procedure in Section 2.8 before Day 3.
60–75%	Acceptable. Log weak sub-topics (closure vs. 2NF vs. 3NF vs. HNF procedure) in the error log below.
75–85%	Good. Log individual misses, then proceed to Day 3 as scheduled.
>85%	Strong. Proceed to Day 3 (BCNF, lossless/dependency preservation in depth, mixed DBMS MCQs) with confidence.

Note: This is a personal study diagnostic only, not an official IOCL cutoff or scoring benchmark. No IOCL-specific subject weightage is implied by these bands.

12 Revision Checklist

12-Item Revision Checklist – tick before moving to Day 3

- ☐ I can state what $X \to Y$ means in plain language.
- ☐ I can distinguish trivial vs. non-trivial FDs by checking $Y \subseteq X$.
- ☐ I can compute a small closure X^+ step-by-step without missing a pass.
- ☐ I know: $X^+ =$ all attributes $\Rightarrow X$ is a super key.
- ☐ I can check minimality to confirm whether a super key is a candidate key.
- ☐ I know a relation can have multiple candidate keys.
- ☐ I can identify prime vs. non-prime attributes from the candidate key set.
- ☐ I can name and explain all three anomalies: update, insertion, deletion.
- ☐ I know 1NF only guarantees atomicity, nothing else.
- ☐ I can identify a partial dependency (2NF issue) from a composite-key example.
- ☐ I know 3NF requires no non-prime attribute transitively depending on the key via another non-prime attribute.
- ☐ I can apply the four-step Highest-Normal-Form procedure (Section 2.8) directly to a relation + key + FD set, and I understand losslessness and dependency preservation are two separate, independently-checked properties.

Things I Got Wrong Today

When logging a mistake below, first tag which category it falls into – this makes revision far more targeted than a generic "got it wrong":

- ☐ FD definition mistake (trivial vs. non-trivial, determinant vs. dependent)
- ☐ Closure calculation mistake (stopped a pass too early, missed a re-scan)
- ☐ Candidate-key minimality mistake (accepted a non-minimal super key as a candidate key)
- ☐ Prime/non-prime mistake (misclassified an attribute)
- ☐ 1NF mistake (missed a non-atomic value)
- ☐ Partial-dependency / 2NF mistake (applied 2NF check to a single-attribute key, or missed that a determinant must be a *proper subset* of a composite key)
- ☐ Transitive-dependency / 3NF mistake (confused it with a partial dependency)
- ☐ Lossless-decomposition mistake (assumed losslessness without checking the common attribute is a key)
- ☐ Dependency-preservation mistake (confused it with losslessness, or assumed one implies the other)

Question	**Topic / Category**	**My mistake**	**Correct rule**	**Revisit date**

13 One-Page Cheat Sheet

FD Notation
$X \to Y$: "X determines Y."
Trivial: $Y \subseteq X$. Non-trivial: $Y \not\subseteq X$.
X = determinant, Y = dependent.

Closure Algorithm
1. $result = X$.
2. If $\alpha \subseteq result$ for some $\alpha \to \beta$, add β.
3. Repeat until no change.
4. $result = X^+$.

Super Key vs. Candidate Key
Super key: X^+ = all attributes.
Candidate key: minimal super key (no attribute removable).
A relation may have several candidate keys.

Prime vs. Non-Prime
Prime: in ≥ 1 candidate key.
Non-prime: in NO candidate key.

1NF / 2NF / 3NF
1NF: atomic values only.
2NF: 1NF + no non-prime attribute partially dependent on part of a composite candidate key.
3NF: 2NF + for every non-trivial $X \to A$, X is a super key OR A is prime.

Partial vs. Transitive
Partial: non-prime depends on a *proper subset* of a composite key.
Transitive: non-prime depends on the key *via another non-prime* attribute.

Highest-Normal-Form Procedure
1. Atomic values? $\to$ else stuck below 1NF.
2. Find candidate key(s); mark prime/non-prime.
3. Composite key + non-prime depends on part of it? $\to$ stuck at 1NF.
4. Non-prime depends on a non-superkey (via another non-prime)? $\to$ stuck at 2NF.
5. Else: 3NF holds.

Lossless vs. Dependency-Preserving
Lossless: rejoining parts = exact original relation (common attribute must be a key in one piece).
Dependency-preserving: original FDs enforceable from the parts alone, without rejoining.
These are INDEPENDENT properties.

High-Yield Traps
1. Super key $\neq$ candidate key (minimality matters).
2. Multiple candidate keys are possible.
3. 2NF issues need a *composite* candidate key.
4. A *prime* attribute on part of a key does NOT break 2NF.
5. Partial $\neq$ transitive dependency.
6. 3NF $\neq$ "zero redundancy."
7. Decompositions are not automatically lossless.
8. Losslessness $\neq$ dependency preservation.
9. BCNF, 4NF, 5NF, Armstrong's axioms, minimal cover – not today (Section 3).

End of Day 2 Workbook (Updated) – IOCL Grade-A 2026 CS/IT Preparation
Next: Day 3 – BCNF, lossless/dependency preservation in depth, mixed DBMS MCQs